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STAT3375Q HW4

I learned L^AT_EX

$$\mathbf{3.103} \quad \frac{\binom{6}{5}}{\binom{10}{5}} \approx 0.0238$$

$$\mathbf{3.106} \quad p(\text{Defective}) = \frac{4}{10}$$

$$E(Y) = np = \$50 \frac{4}{10} = \$20$$

$$Var(Y) = np(1-p) = 20(.6) = \$12$$

3.105 **a** Hypergeometric. We are selecting $n = 3$ randomly sampled objects from a population of size $N = 8$ with $r = 5$ Class 1 objects.

$$\mathbf{b} \quad P(Y \geq 2) = P(Y = 2) + P(Y = 3)$$

$$= \frac{\binom{5}{2} \binom{8-5}{3-2}}{\binom{8}{3}} + \frac{\binom{5}{3} \binom{8-5}{3-3}}{\binom{8}{3}} \approx .714$$

$$\mathbf{c} \quad E(Y) = n \frac{r}{N} = 3 \frac{5}{8} = 1.875$$

$$Var(Y) = n \left(\frac{r}{N} \right) \left(\frac{N-r}{N} \right) \left(\frac{N-n}{N-1} \right) = 3 \frac{5}{8} \frac{8-5}{8} \frac{8-3}{8-1} \approx .502$$

3.107 $N = 10, n = 4, r = 5$

$$\mathbf{a} \quad P(Y = 4) = \frac{\binom{5}{4} \binom{10-5}{4-4}}{\binom{10}{4}} \approx .0238$$

$$\mathbf{b} \quad P(Y \leq 3) = 1 - P(Y = 4) \approx .976$$

$$\mathbf{c} \quad P(Y \geq 1) = 1 - P(Y = 0) \approx .976$$

3.113 $N = 20, n = 6, r = 8$

$$P(Y = 1) = \frac{\binom{8}{1} \binom{12}{5}}{\binom{20}{6}} \approx .163, \text{ which is not negligible.}$$

$$\mathbf{3.114} \quad E(Y) = 6 \frac{8}{20} = 2.4, Var(Y) = 6 \frac{8}{20} \frac{20-8}{20} \frac{20-6}{19} \approx 1.06$$

$$\mathbf{3.118} \quad \frac{P(Y=4)}{P(Y=3)} \quad N = 52, n = 5, r = 4$$

$$\frac{\binom{4}{4} \binom{48}{1}}{\binom{52}{5}} \approx .0106$$

$$\mathbf{3.121} \quad \mathbf{a} \quad P(Y = 4) = \frac{e^{-2} 2^4}{4!} = \frac{2}{3} e^{-2}$$

$$\mathbf{b} \quad P(Y \geq 4) = 1 - P(Y < 4) = P(Y = 0) + P(Y = 1) + P(Y = 2) + P(Y = 3) \approx 1 - .1353 + .2707 + .2707 + .1804 \approx .143$$

$$\mathbf{c} \quad P(Y < 4) = 1 - P(Y \geq 4) = 1 - .143 = .857$$

$$\mathbf{d} \quad \frac{P(Y \geq 4)}{1 - P(Y < 2)} = \frac{.143}{1 - .135 + .271} \approx .241$$

$$\mathbf{3.123} \quad \frac{e^{-\lambda} \lambda^0}{0!} = \frac{e^{-\lambda} \lambda^1}{1!}$$

$$\lambda = 1, p(2) \approx .184$$

$$\mathbf{3.127} \quad \lambda = 4, P(Y \leq 4) \approx .0183 + .0733 + .1465 + .1954 + .1954 \approx .629$$

$$\mathbf{3.139} \quad E(Y) = 50 - 2(2) - 2^2 = 42$$

$$\mathbf{10} \quad P(a = 0) = \frac{e^{-1.3*4} (1.3*4)^0}{0!} \approx .0055$$

$$P(b = 5) = \frac{\binom{10}{5} \binom{18-10}{5-5}}{\binom{18}{5}} \approx .0294$$

Therefore it is more likely that Martha spent the whole night with ghosts.